

Mahsa Amani

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Research Interests

- 📖 **Agentic AI**
- 📖 **Large Language Models**
- 📖 **Natural Language Processing**

Education

Max Planck Institute for Software Systems (MPI-SWS)

Saarbrücken, Germany

PH.D. IN COMPUTER SCIENCE

Jun. 2024 - Present

- Supervised by Prof. Krishna P. Gummadi

Sharif University of Technology (SUT)

Tehran, Iran

B.SC. IN COMPUTER ENGINEERING

Sep. 2018 - Feb. 2023

- GPA: 18.03/20

Farzanegan High School (National Organization for Development of Exceptional Talents (NODET))

Tabriz, Iran

DIPLOMA IN MATHEMATICS AND PHYSICS

Sep. 2011 - Jun. 2018

- GPA: 19.97/20

Academic Honors

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|-----------|-----------------------------|---|------|
| 2022 | Gifted Student Award | Entrance exam waiver for graduate program, Sharif University of Technology | Iran |
| 2018-2023 | Membership | In Iran's National Elites Foundation | Iran |
| 2018 | Rank 51st | In Mathematics and Physics B.Sc. entrance exam among more than 144,000 participants | Iran |
| 2017 | Accepted | In the First Exam of Mathematics Olympiad | Iran |

Publications

Characterizing Web Search by Conversational LLM Agents: From Search Decisions and Strategies to Results and Responses

M. AMANI, S. LEE, , A. DASH, A. EL FRAIHI, Y. JANG, E. KIRSTEN, Q. WU, K. P. GUMMADI, M. GUPTA, A. RAVICHANDER, S. DAS

2026

- ArXiv

BLOCK: A Relative Representational Geometry for Understanding Large Language Models

M. AMANI, S. DAS, Q. WU, M. A. KHAN, S. LEE, K. P. GUMMADI, M. B. ZAFAR

2026

- In progress

To Call or Not to Call: A Framework to Assess and Optimize LLM Tool Calling

Q. WU, S. DAS, M. AMANI, A. NAG, S. LEE, K. P. GUMMADI, A. RAVICHANDER, M. B. ZAFAR

2026

- ArXiv

In Agents We Trust, but Who Do Agents Trust? Latent Source Preferences Steer LLM Generations

M. A. KHAN, M. AMANI, S. DAS, B. GHOSH, Q. WU, K. P. GUMMADI, M. GUPTA, A. RAVICHANDER

2026

- The Fourteenth International Conference on Learning Representations (ICLR)

Rote Learning Considered Useful: Generalizing over Memorized Data in LLMs

Q. WU, S. DAS, M. AMANI, B. GHOSH, M. A. KHAN, K. P. GUMMADI, M. B. ZAFAR

2026

- The Fourteenth International Conference on Learning Representations (ICLR)

Revisiting Privacy, Utility, and Efficiency Trade-offs when Fine-Tuning Large Language Models

S. DAS, C. KOLLING, M. A. KHAN, M. AMANI, B. GHOSH, Q. WU, T. SPEICHER, K. P. GUMMADI

2025

- ArXiv

Linguistic Resources and Transformer-based Models for the Machine Translation

M. AMANI, M. NOURI, Z. BAHMANI, M. MIRBEYGI, N. HASHEMI, E. ASGARI, M. SOLEYMANI BAGHSHAH, H. BEIGY, A. MOVAGHAR

2024

- Journal of the Linguistics Society of Iran

The Language Model, Resources, and Computational Pipelines for the Under-Resourced Iranian Azerbaijani

M. AMANI, M. NOURI, R. ZOHRABI, E. ASGARI

2023

- In Proceedings of AACL 2023 (Asia-Pacific ACL)

Attended Conferences and Workshops

2025	Association for Computational Linguistics, Vienna, Austria	July 2025
2025	ACM International Conference on Web Search and Data Mining, Hannover, Germany	March, 2025

Projects

Unveiling Hospital Cost Drivers	Germany
EXPLAINABLE AND AGENTIC AI APPROACH FOR UNVEILING HOSPITAL COST DRIVERS	Mar. 2025 - August. 2025
3D Vision Transformers and Convolutional Models for Segmentation Tasks	Iran
BACHELOR’S THESIS ON 3D MRI BRAIN TUMOR SEGMENTATION	Jan. 2022 - Jan. 2023
Iranian Azerbaijani NLP Pipeline	Iran
THE LANGUAGE MODEL, RESOURCES, AND COMPUTATIONAL PIPELINES FOR THE UNDER-RESOURCED IRANIAN AZERBAIJANI	Jan. 2023 - May. 2023
Turkish Preprocessing Pipeline	Iran
DEVELOPING A PREPROCESSING PIPELINE FOR TURKISH LANGUAGE	April 2022
Eye Disease Recognition	Iran
RECOGNITION OF 7 EYE DISEASES USING TRANSFER LEARNING IMPLEMENTED AS MY INTERNSHIP PROJECT	Sep. 2021

Research Experience

Pipelines for low-resource Iranian Languages	Iran
RESEARCH ASSISTANT	Sep. 2022 - Dec. 2023
• Developing pipelines for low-resource languages such as Azerbaijani, Luri, Yazdi, and Kurdi	
A Large Language Model for Persian	Iran
RESEARCH ASSISTANT	Oct. 2023
• Development of a multilingual large language model fine-tuned for Persian	

Work Experience

Teaching Assistant	Saarland, Germany
MAX PLANCK INSTITUTE	Oct. 2025 - Feb. 2026
• Efficient Training of Large Language Models: From Basics to Fine-Tuning , Prof. Krishna Gummadi	
Junior Data Scientist	Tehran, Iran
ZAREBIN	Oct. 2022 - Apr. 2023
• Developing a Shopping Information Retrieval System using Deep Learning Methods	
Teaching Assistant	Tehran, Iran
SHARIF UNIVERSITY OF TECHNOLOGY (SUT)	Sep. 2020 - Jul. 2023
• Natural Language Processing, Dr. Ehsaneddin Asgari	
• Probability and Statistics, Dr. Ali Sharifi	
• Database Design, Dr. Abbas Heydarnoori	
• Computer Architecture, Dr. Hossein Asadi	
• Python Programming, Dr. Issa Moradnejad	
Undergraduate Student Internship	Tehran, Iran
AIMEDIC	Jul. 2021 - Sep. 2021
• Eye Disease Diagnosis using Artificial Intelligence	

Skills

Programming </>	Python, R, C/C++, JAVA, SQL, L ^A T _E X
ML tools 🛠️	HuggingFace, VLLM, Pytorch, Tensorflow, Keras, Scikit-learn, Numpy, Pandas

Languages

English	IELTS Overall Bandscore: 7.5
German	Elementary (A1)
Persian	Native Language
Azerbaijani	Mother Tongue